

From Time–Frequency Localization to Risk-Sensitive Access: A Belief-Map Closed Loop for Wideband Cognitive Radio

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Abstract—Dynamic spectrum access (DSA) schemes based on deep reinforcement learning (DRL) almost universally assume that the wideband spectrum is pre-partitioned into a fixed grid of channels, each in a binary good/bad state. This assumption breaks down in real wideband environments, where primary-user (PU) emissions occupy *arbitrary* center frequencies, bandwidths, and durations—precisely the regime addressed by time–frequency (TF) localization. We close the loop between the two: the sensing module outputs a *continuous occupancy belief map* over the TF plane, which serves directly as the state of the access agent; the agent selects a TF resource block. We cast the problem as a partially observed control problem with a learned soft-occupancy observation, and analyze (i) a propagation bound mapping localization error ($P_{\text{md}}, P_{\text{fa}}$) to PU-collision probability and throughput loss, yielding a three-way sensing-accuracy/throughput/PU-protection trade-off with a characterized operating point; (ii) an information-refinement result showing the belief-state representation is no worse than a coarser fixed-channel one under a common action space; and (iii) a risk-sensitive (CVaR) access formulation whose operating point is provably at least as conservative as the mean-constrained one. A stylized regret analysis for a tabular surrogate separates a vanishing learning cost from a policy-independent sensing gap. Simulations on a continuous-bandwidth, non-grid PU model confirm the predicted operating point and the tail-risk reduction, and show that access performance is governed by *frequency-localization accuracy*—which the analysis ties to access throughput through the localizer’s IoU.

Index Terms—Dynamic spectrum access, spectrum sensing, time–frequency localization, deep reinforcement learning, partial observation, cognitive radio.

I. INTRODUCTION

SPECTRUM scarcity has become a defining bottleneck for modern and forthcoming wireless systems. Dynamic spectrum access (DSA), in which secondary users (SUs) opportunistically reuse spectrum left idle by primary users (PUs), is widely regarded as a key remedy and underpins emerging shared-spectrum regimes such as the citizen broadband radio service (CBRS) [1]. Realizing DSA requires solving two tightly coupled subproblems: *spectrum sensing*, which estimates what spectrum is free, and *spectrum access*, which decides how to use it. Because the quality of sensing directly bounds what access can achieve, and because the performance after access is the only meaningful measure of whether a sensing decision was good, a joint treatment of the two is necessary rather than optional.

Two largely separate research lines have matured around these subproblems.

The first, *time–frequency (TF) localization* (TFL), treats the wideband spectrogram as an image and applies object detection to output, for every emission, a bounding box describing its center frequency, bandwidth, start time, and duration [2], [3]. Because the wideband spectrogram and image object detection are structurally similar, a sequence of detectors has been adapted to TFL, including region-based and YOLO-family architectures [2], [3], anchor-free keypoint detectors that resolve the prior-anchor mismatch caused by the diverse aspect ratios of RF emissions [4], and transformer backbones that remain robust when emissions overlap in the TF plane [5], [6]. This line has steadily improved how precisely and robustly a wideband sensor can describe occupancy. Yet it terminates at sensing: it produces a rich, continuous, often uncertainty-bearing description of the spectrum and then does nothing with it. The access decision is left out of scope.

The second line applies *deep reinforcement learning (DRL) to DSA*, learning access—and sometimes sensing—policies under partial observability [7]–[12]. Deep Q -networks [7], actor–critic methods [8], recurrent Q -networks for partial observation [9], [13], and channel-aggregation schemes [10] have progressively enlarged the action space and improved sample efficiency. Almost without exception, however, this line assumes the wideband spectrum is pre-partitioned into a *fixed grid of channels*, each summarized by a binary good/bad or idle/busy state.

These two lines are separated by an assumption that is rarely examined. The DRL-DSA line inherits a fixed channelization in which PU activity is aligned to grid cells. But real wideband emissions occupy *arbitrary* center frequencies, bandwidths, and durations—and in shared-spectrum regimes such as CBRS the center frequency and subchannel division of incumbents change adaptively with traffic and channel state [1], [5]. This is precisely the regime the TFL line was built to handle. Once occupancy is not grid-aligned, the “good/bad channel” state that DRL-DSA agents consume is itself a lossy, and sometimes incorrect, summary of the environment: a single grid cell may straddle the boundary of a PU emission, forcing the agent either to collide or to abandon usable spectrum. To our knowledge, no prior work feeds the *continuous, uncertainty-bearing* TF occupancy estimate produced by a localizer *into* the access decision as the agent’s state, nor analyzes how the localizer’s errors propagate to access performance and PU protection. Closing that loop, and analyzing it, is the subject

of this paper.

We propose *Localization-Aware Joint Spectrum Sensing and Access* (LA-JSSA), in which the sensing module emits a continuous occupancy *belief map* over the TF plane that directly serves as the access agent's state, and the agent selects a continuous TF resource block for transmission. We further develop a distributional [14], risk-sensitive access tier (D-LA-JSSA) tailored to the belief-state representation. Our contributions are:

- 1) **A belief-map closed loop.** We formulate joint sensing and access as a partially observed control problem whose state is the continuous TF occupancy posterior rather than a fixed-channel vector (Section III), realized by a two-timescale architecture that couples a supervised TFL sensing tier to a reinforcement-learned access tier without unstable end-to-end training (Section IV).
- 2) **Error-propagation analysis and a three-way trade-off.** We map cell-level localization error ($P_{\text{md}}, P_{\text{fa}}$) to PU-collision probability and SU throughput, yielding a sensing-accuracy/throughput/PU-protection trade-off with a characterized optimal belief threshold η^* (Theorem 1), and a linear bound on the price of imperfect localization (Corollary 1).
- 3) **Risk-sensitive access via a distributional critic.** We cast access as a *distributional* RL problem [14], [15] with a tail-risk (CVaR) constraint on PU collision, and show its operating point is at least as conservative as the mean-constrained one, with a first-order estimate of the conservativeness margin (Theorem 2, Corollary 3).
- 4) **Information refinement and a stylized regret analysis.** We show the belief-state policy class is no worse than any coarser fixed-channel representation under a common action space (Proposition 1), with a strict gap on a misaligned instance (Example 1); and, for a tractable tabular surrogate, we separate the regret into a vanishing learning term and a policy-independent sensing gap (Proposition 2). The latter is a stylized analysis, not a guarantee for the deep agent.
- 5) **Validation.** Simulations on a continuous-bandwidth, non-grid PU model confirm the predicted operating point and the tail-risk reduction (Section VI).

The remainder of the paper is organized as follows. Section II reviews related work, Section III formulates the model, Section IV presents LA-JSSA, Section V develops the theory, Section VI reports experiments, and Section VII concludes.

II. RELATED WORK

We organize prior work into three strands—TF localization, DRL-based access, and joint sensing-and-access—and state, for each, the precise distinction from LA-JSSA.

A. Time–Frequency Localization for Wideband Sensing

Classical spectrum sensing—energy, matched-filter, cyclostationary, and covariance-based detectors—is surveyed in [16]–[18]. Scanning-based wideband sensing divides the spectrum into subbands according to a known channelization and applies narrowband detection in each subband [7]. In

shared bands such as CBRS and ISM, however, the center frequency and subchannel division of incumbents change adaptively with service demand and channel state, so methods that presuppose a fixed channelization fail [1]. Deep learning has since been applied to over-the-air signal classification [19], [20] and wideband signal detection [21]. TFL was introduced precisely to remove the fixed-channelization assumption: by casting the magnitude spectrogram as an image and applying object detection, a detector can simultaneously estimate the time and frequency support of every emission [2], [22].

A detector lineage has followed. Downscaled Faster-RCNN and YOLO-family detectors [23] were first adapted to signal TFL [2], [3]. The diverse aspect ratios of RF emissions cause a prior-anchor mismatch in anchor-based detectors; anchor-free keypoint detectors with tailored matching were proposed to resolve it and to classify multiple co-existing emitters [4]. When emissions overlap in the TF plane—as they do under concurrent spectrum reuse—convolutional backbones lose discriminative features; transformer backbones that exploit TF correlation, optionally with masked-autoencoder pre-training [24] to overcome label scarcity, achieve state-of-the-art robustness [5], [6], [24]. Collectively, this line has made the occupancy description ever more precise and robust. *Our distinction:* every one of these works terminates at sensing, emitting boxes (or, after non-maximum suppression, hard decisions) that are never coupled to an access policy. We instead retain the detector's *soft* output as a belief state and close the loop to access.

B. DRL for Dynamic Spectrum Access

DRL-based DSA learns channel-selection policies under partial observability, building on the deep Q -network [25] and modern policy-gradient methods such as PPO [26] and soft actor–critic [27]. Deep Q -networks were applied to multichannel access with correlated channel occupancy [7]; actor–critic frameworks improved performance in large action spaces [8]; recurrent Q -networks addressed partial observation by summarizing observation histories [9], [13], [28]; reservoir-computing and multi-user variants targeted distributed access [29], [30]; UCB-style exploration was used in cooperative sensing [31]; and spectrum-aggregation schemes let an SU combine several idle channels to meet a bandwidth demand [10]. Related efforts apply DRL to joint channel and power control [32] and to cognitive radio sensor networks [33]. These methods share a structural premise: the state is defined on a *fixed channel grid* with binary occupancy, and the action selects one or more whole grid channels. *Our distinction:* our state is a continuous TF belief map with no grid, our action is a continuous resource block, and—uniquely—we analyze how sensing error propagates into this continuous access problem.

Safety and risk in RL are also relevant to our CVaR access constraint. Constrained MDPs [34] and their deep solvers (constrained policy optimization [35], reward-constrained policy optimization [36]) enforce expected-cost limits, while percentile- and CVaR-based criteria [37], [38] and distributional RL [14], [15], [39] target the tail. We adopt a distributional critic with a CVaR collision constraint and connect its operating point to the sensing error.

TABLE I
SUMMARY OF NOTATION

Symbol	Meaning
$\mathbf{X} \in \mathbb{R}_{\geq 0}^{N_f \times N_t}$	Magnitude spectrogram (frame)
(m, n)	TF cell (freq. bin m , time bin n)
$\mathbf{b}_k = (f_k, \beta_k, \tau_k, \delta_k)$	k -th PU emission box
$\mathbf{O} \in \{0, 1\}^{N_f \times N_t}$	True occupancy map
$\mathbf{B} \in [0, 1]^{N_f \times N_t}$	Estimated occupancy belief map
$P_{\text{md}}, P_{\text{fa}}$	Cell-level miss/false-alarm prob.
η	Belief decision threshold
$\mathbf{a} = (f^a, \beta^a, \tau^a, \delta^a, p)$	SU access action (resource block)
$\beta_{\text{min}}, p_{\text{max}}$	Min. bandwidth demand, power budget
$\mathbf{s}_t, \mathbf{o}_t$	Access state, observation
$\mathcal{R}, \gamma, \pi$	Reward, discount, policy
$P_{\text{col}}, P_{\text{col}}^{\text{max}}$	PU collision prob. and its cap
$\mathbf{R}, \mathbf{R}_{\text{max}}$	Achievable / oracle throughput

C. Joint Sensing-and-Access

Closest to us are works that jointly design sensing and access. The classical sensing–throughput trade-off optimizes the sensing *duration* and an energy threshold for a *single* channel, trading detection against false alarm along the time axis [40]; it does not address a two-dimensional continuous TF plane or a learned localizer. Among learning approaches, double deep Q -learning has been used to learn sensing and access policies jointly, but in a *narrowband, single-channel* setting without TF localization or continuous boxes [12]. Hierarchical multi-agent DRL jointly selects a sensing window and power allocation, yet its state remains discrete band occupancy on a fixed grid [11]. Cooperative sensing-and-access for industrial IoT integrates the two via optimization rather than learning [41]. Interference-temperature-constrained DSA formalizes PU protection but again on predefined channels [42]. *Our distinction:* LA-JSSA operates on a two-dimensional continuous TF plane, drives access directly from a learned occupancy belief map, analyzes the resulting sensing-accuracy/throughput/PU-protection trade-off as a function of localization error, and protects the PU through a tail-risk (CVaR) constraint rather than a mean constraint—a regime, to our knowledge, not previously treated.

III. SYSTEM MODEL AND PROBLEM FORMULATION

We consider a single secondary user (SU) opportunistically accessing a wideband spectrum of bandwidth W centered at f_c , which is shared by an *unknown and time-varying* number of primary-user (PU) emissions. Such a setting arises in shared bands such as CBRS, where incumbents select center frequencies and bandwidths adaptively rather than on a fixed channel grid [1], [42]. Unlike the fixed-channel models prevalent in the DRL-based DSA literature, we make *no* assumption that PU activity is aligned to a predefined channel grid: each PU emission occupies an arbitrary center frequency, bandwidth, start time, and duration, exactly the regime targeted by TF localization [2], [5]. The notation introduced below is summarized in Table I.

A. Wideband Signal and Spectrogram Model

The complex baseband signal observed by the wideband sensor over one frame is

$$r(t) = \sum_{k=1}^{N_p} c_k(x_k(t)) + w(t), \quad 0 \leq t < T_f, \quad (1)$$

where $x_k(t)$ is the k -th PU emission, $c_k(\cdot)$ denotes the corresponding channel (carrying carrier offset, bandwidth-limiting, and fading), $w(t)$ is additive white Gaussian noise, N_p is the (random, unknown) number of active emissions in the frame, and T_f is the frame length. The signal is sampled at rate $f_s \geq W$.

Applying the short-time Fourier transform (STFT) operator $\mathcal{S}$ yields a single-resolution time–frequency (TF) representation; we work with the magnitude spectrogram

$$\mathbf{X} = |\mathcal{S}\{r\}| \in \mathbb{R}_{\geq 0}^{N_f \times N_t}, \quad (2)$$

discretized into N_f frequency bins and N_t time bins. We index an individual TF *cell* by (m, n) , $m \in \{1, \dots, N_f\}$, $n \in \{1, \dots, N_t\}$. The STFT is preferred over variable-resolution transforms (CWT/CQT) because spectrum monitoring weights all frequencies equally [2].

B. Continuous Occupancy Representation

Each PU emission k is described by a continuous TF *box*

$$\mathbf{b}_k = (f_k, \beta_k, \tau_k, \delta_k), \quad (3)$$

i.e. center frequency, bandwidth, start time, and duration. The box maps to a region $\mathcal{B}_k \subset \{1, \dots, N_f\} \times \{1, \dots, N_t\}$ of the discretized TF plane. The *true occupancy map* is the binary matrix $\mathbf{O} \in \{0, 1\}^{N_f \times N_t}$ with

$$O_{m,n} = 1\left\{ (m, n) \in \bigcup_{k=1}^{N_p} \mathcal{B}_k \right\}, \quad (4)$$

where $O_{m,n} = 1$ marks a cell occupied by at least one PU. Crucially, $\mathbf{O}$ is *not* block-structured along a fixed channelization; recovering it is exactly the TF-localization problem.

We clarify the sense in which the model is “grid-free.” The PU *emission parameters* (3) are continuous; they are mapped to an occupancy map discretized at the *sensing resolution* $N_f \times N_t$ of the STFT. The point is not the absence of any discretization—a finite spectrogram is necessarily discretized—but the absence of a fixed *communication channel grid* to which PU activity is assumed aligned. Emissions may occupy arbitrary cells at sensing resolution, straddling any notional channel boundary, which is precisely what defeats fixed-channel DSA models and motivates the localization-driven treatment.

C. Sensing Model: From Spectrogram to Belief Map

The sensing module is a learned mapping ϕ_{sen} that estimates, for each cell, the posterior probability of PU occupancy:

$$\mathbf{B} = \phi_{\text{sen}}(\mathbf{X}) \in [0, 1]^{N_f \times N_t}, \quad B_{m,n} \approx \mathbb{P}(O_{m,n}=1 \mid \mathbf{X}). \quad (5)$$

Concretely, $B_{m,n}$ is the *detection confidence* of the time–frequency localizer rasterized onto the cell grid: each estimated

emission box carries a confidence score, and $B_{m,n}$ is the (edge-feathered) score of the box covering cell (m,n) , or a low background value where no box is predicted. We deliberately retain this *soft* confidence $\mathbf{B}$ rather than applying non-maximum suppression to obtain hard boxes; this preserves the sensing uncertainty that the access stage will exploit (cf. Section IV). The reliability of ϕ_{sen} is characterized at the cell level by the miss-detection and false-alarm probabilities

$$P_{\text{md}} \triangleq \mathbb{P}(B_{m,n} < \eta \mid O_{m,n} = 1), \quad (6)$$

$$P_{\text{fa}} \triangleq \mathbb{P}(B_{m,n} \geq \eta \mid O_{m,n} = 0), \quad (7)$$

where $\eta \in [0, 1]$ is a decision threshold applied to the localizer's confidence $B_{m,n}$. Two distinct factors shape $(P_{\text{md}}, P_{\text{fa}})$: the *confidence threshold* η (where we cut the soft score) and the *geometric accuracy* of the box (how well its frequency support matches the true emission, quantified later by the IoU error ε_{loc}). The operating-point analysis of Theorem 1 optimizes over η , while the localization-error model of Assumption 1 captures the geometric contribution; their separation is made explicit in the bound (16). Both P_{md} and P_{fa} are functions of η ; this dependence is the lever in the trade-off of Theorem 1 (Section V).

D. Access Model: Continuous Resource Selection

The SU operates a sense-then-access frame: the sensing phase estimates the occupied frequency ranges, and the access phase of the *same* frame transmits in the ranges it deems idle. Because both phases share the frame, temporal alignment is given, and the access decision turns on *which frequencies* are free—making frequency-localization accuracy, rather than temporal extent, the quantity that governs access performance (formalized in Section V). At each frame the SU chooses a continuous TF resource block together with a transmit power level,

$$\mathbf{a} = (f^a, \beta^a, \tau^a, \delta^a, p), \quad (8)$$

subject to the bandwidth demand, power, and PU-protection constraints

$$\beta^a \geq \beta_{\text{min}}, \quad 0 \leq p \leq p_{\text{max}}, \quad \mathbb{P}(\text{collision}) \leq P_{\text{col}}^{\text{max}}. \quad (9)$$

A *collision* occurs when the SU block overlaps an occupied cell:

$$\text{Col}(\mathbf{a}) \triangleq \mathbf{1}\{\exists (m,n) \in \mathcal{A}(\mathbf{a}) : O_{m,n} = 1\}, \quad (10)$$

where $\mathcal{A}(\mathbf{a})$ is the set of TF cells covered by action $\mathbf{a}$. Transmission succeeds (yielding an ACK) iff $\text{Col}(\mathbf{a}) = 0$ and the allocated bandwidth meets the demand.

E. Partially Observed Control Formulation

The SU never observes the true occupancy $\mathbf{O}$ directly; it observes only the wideband spectrogram $\mathbf{X}_t$ and, after transmitting, a one-bit ACK/NACK. The control problem is therefore partially observed. We make the roles explicit to avoid conflating the detector output with a Bayesian belief.

Hidden state: The hidden state is the true PU occupancy together with the underlying emission parameters, $\mathbf{s}_t^h = (\mathbf{O}_t, \{\mathbf{b}\})$, evolving as a (piecewise-stationary) Markov process.

Observation: The raw observation is the spectrogram $\mathbf{X}_t$ and the ACK/NACK $\text{ACK}_t \in \{0, 1\}$. The sensing tier maps $\mathbf{X}_t$ to the per-cell soft occupancy estimate $\mathbf{B}_t \in [0, 1]^{N_f \times N_t}$ of (5). We stress that $\mathbf{B}_t$ is a learned *soft occupancy observation feature*, not the exact Bayesian posterior $\mathbb{P}(\mathbf{s}_t^h \mid \text{history})$; it plays the role of an *approximate sufficient statistic* of the observation history for the access decision.

Access state: The access policy acts on the recent history of this feature,

$$\mathbf{s}_t = (\mathbf{B}_{t-H+1}, \dots, \mathbf{B}_t) \in [0, 1]^{H \times N_f \times N_t}, \quad (11)$$

stacked over a short window to capture temporal correlation of PU activity. This continuous, uncertainty-bearing state is the central departure from prior DRL-DSA work, whose state is a binary fixed-channel occupancy vector. Because $\mathbf{B}$ approximates rather than equals the exact posterior, the formulation is a *partially observed control problem with a learned observation feature* rather than a belief-MDP in the strict Bayesian sense; the theoretical claims of Section V are stated against this representation and do not presume exact belief recursion.

Reward: We use an uncertainty-aware reward that rewards throughput, penalizes PU collisions, and discourages transmitting into low-confidence regions:

$$\mathcal{R}(\mathbf{s}_t, \mathbf{a}_t) = \underbrace{\mathcal{R}(\mathbf{a}_t)}_{\text{throughput}} - \lambda_c \text{Col}(\mathbf{a}_t) - \lambda_u \mathcal{U}(\mathbf{B}_t, \mathbf{a}_t), \quad (12)$$

where $\mathcal{U}(\cdot)$ measures the residual occupancy uncertainty over the chosen block and $\lambda_c, \lambda_u \geq 0$ are weights. The connection of the λ_u term to a risk-sensitive objective is discussed in Section IV.

Objective: The SU seeks a policy $\pi : \mathcal{S} \rightarrow \mathcal{A}$ maximizing the expected discounted return subject to the PU-protection constraint,

$$\pi^* = \arg \max_{\pi} \mathbb{E}_{\pi} \left[\sum_{t=0}^{\infty} \gamma^t \mathcal{R}(\mathbf{s}_t, \mathbf{a}_t) \right] \quad (13)$$

s.t. $\mathbb{E}_{\pi}[\text{Col}(\mathbf{a}_t)] \leq P_{\text{col}}^{\text{max}}.$

Remark 1. Setting $\mathbf{B}$ to a binary fixed-channel vector and restricting $\mathbf{a}$ to a single grid channel recovers the standard formulation of [7], [8] as a special case. Proposition 1 formalizes that this restriction can be strictly suboptimal under grid misalignment.

IV. THE LA-JSSA FRAMEWORK

We now describe the two-timescale architecture that realizes the partially observed control problem of Section III. The design separates a slowly-adapted supervised *sensing tier*, which produces the occupancy belief map $\mathbf{B}$, from a fast *access tier* that learns the policy π by reinforcement [43]. Fig. 1 shows how the two tiers close the sensing-access loop: the data flow $r \rightarrow \mathbf{X} \rightarrow \mathbf{B} \rightarrow \mathbf{s} \rightarrow \mathbf{a}$ feeds forward through the network, and ACK/collision feedback from the environment returns to both the localizer and the risk-constrained critic.

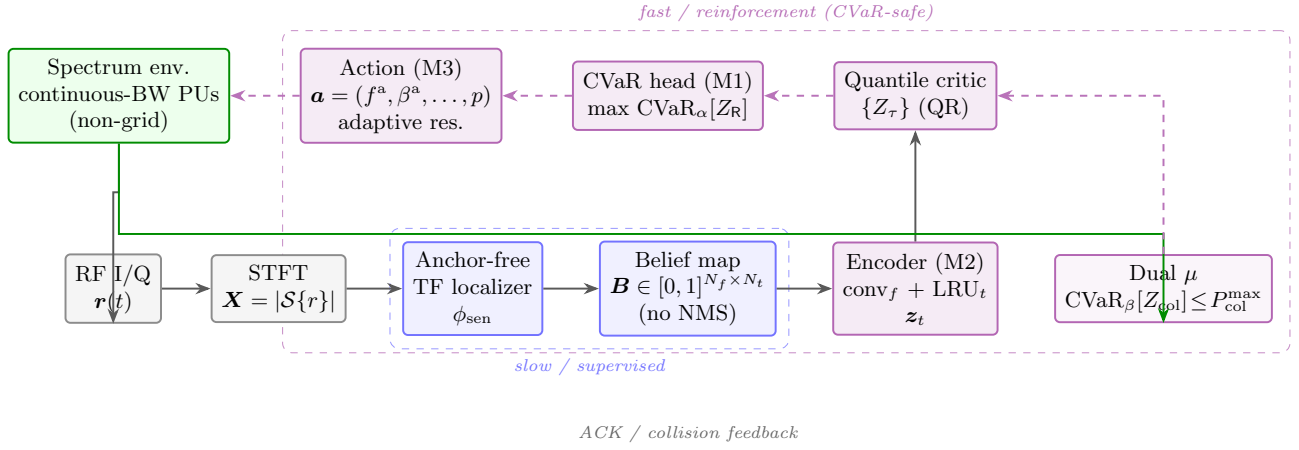


Fig. 1. The D-LA-JSSA two-timescale architecture. *Bottom (slow, supervised)*: wideband I/Q is transformed to a spectrogram and passed to an anchor-free TF localizer ϕ_{sen} that emits a soft occupancy belief map $\mathbf{B}$ (no NMS), encoded across time by a $\text{conv}_f + \text{LRU}_t$ encoder (M2). *Top (fast, reinforcement)*: a quantile critic feeds a CVaR head (M1) and an adaptive-resolution action head (M3); a dual-ascent multiplier enforces the CVaR collision constraint from ACK/collision feedback. The environment of continuous-bandwidth, non-grid-aligned PUs closes the loop.

A. Two-Timescale Rationale

Joint end-to-end training of a detector and an RL agent is notoriously unstable: the non-stationary reward signal corrupts the supervised gradient of the detector, while a drifting detector keeps the RL state distribution non-stationary. We therefore decouple the two. The sensing tier ϕ_{sen} is trained offline (or slowly online) on labeled spectrograms with a supervised objective, so that the cell-level error rates $P_{\text{md}}(\eta)$, $P_{\text{fa}}(\eta)$ of Assumption 2 are stable; the access tier then learns against an approximately stationary observation channel. This mirrors the decoupled offline/online structure that is known to accelerate POMDP learning when the observation model is fixed.

B. Sensing Tier: Belief-Map Generation

The sensing tier maps a spectrogram frame to the per-cell occupancy posterior (5). Because PU emissions have widely varying time–frequency aspect ratios, we adopt an *anchor-free* detector (center/extent regression over the TF plane) rather than an anchor-based region proposal network, which couples poorly to continuous, arbitrary bandwidths [4]; when emissions overlap, a transformer backbone that exploits TF correlation further improves robustness [5]. The key departure from prior TF-localization work is that we do *not* apply non-maximum suppression to collapse the heatmap into hard boxes; instead the soft per-cell confidence is retained as the belief map $\mathbf{B} \in [0, 1]^{N_f \times N_t}$. This preserves exactly the uncertainty that Theorem 1 and the reward (12) consume.

Remark 2 (Design insight). *Discarding the soft confidence (the conventional detector output) is information-lossy in the precise sense of Proposition 1: a hard-box state is a deterministic coarsening of $\mathbf{B}$. The belief map is an informative representation for the downstream access decision.*

C. State Encoding Across Time (M2)

To capture temporal correlation of PU activity under partial observability, the access state (11) stacks the most recent

H belief maps. The belief map is a two-dimensional TF object, so we encode it with a *factored* encoder: a convolution along the frequency axis captures the spatial shape of emissions (their TF extent), while a point-wise linear recurrent unit (LRU) along the time axis captures the long-range correlation of PU on/off dynamics, producing a compact latent state $\mathbf{z}_t = \text{Enc}(\mathbf{s}_t)$. We adopt an LRU rather than an LSTM or a Transformer because recent evidence shows point-wise recurrent state-space models outperform attention on partially observable RL while remaining parallelizable and lightweight [44], [45]. The recurrent encoding plays the role of the belief-update / sufficient-statistic recursion of a POMDP. We ablate this choice (LRU vs. ConvLSTM) in Section VI.

D. Access Tier: Continuous Resource Selection

The access tier maps the latent state to the resource-block action (8). We consider two instantiations:

- a value-based agent over a discretized action grid (deep Q -network [13] on $\mathbf{z}_t$), suitable when the TF action set is coarsely quantized; and
- a continuous-control agent (deterministic policy gradient [46]) that outputs $(f^a, \beta^a, \tau^a, \delta^a, p)$ directly, suitable for fine-grained continuous resource selection.

Both optimize the constrained objective (13); the PU-protection constraint is enforced via a Lagrangian penalty whose multiplier tracks the running collision rate against $P_{\text{col}}^{\text{max}}$.

E. Distributional Safe Access (M1)

Standard value-based agents learn only the expected return $\mathbb{E}[Z]$, which discards exactly the information the belief map carries: the dispersion of outcomes induced by occupancy uncertainty. We instead learn the full return *distribution* via a quantile critic, maintaining quantile estimates of the throughput return Z_R and, separately, of the collision return Z_{col} . Two consequences follow.

First, the access objective becomes *risk-sensitive*: rather than $\max \mathbb{E}[Z_R]$ we maximize a tail-aware utility $\text{CVaR}_\alpha[Z_R]$ [38], which protects against rare low-throughput episodes. Second, and more importantly for PU protection, the constraint is imposed on the *tail* of the collision-window return Z_{col}^W of (23),

$$\max_{\pi} \text{CVaR}_\alpha[Z_R] \quad \text{s.t.} \quad \text{CVaR}_\beta[Z_{\text{col}}^W] \leq P_{\text{col},W}^{\max}, \quad (14)$$

which is the formulation analyzed in Theorem 2. The CVaR terms are estimated from the learned quantiles and the constraint is enforced by dual ascent on a multiplier μ as in Algorithm 1, with the running windowed collision-CVaR estimate in place of the mean. By Theorem 2 the resulting operating point is at least as conservative as the mean-constrained one, by a margin that vanishes as the localizer sharpens—tying the method back to the value of TF localization.

Remark 3 (Adaptive action resolution (M3)). *The quantile critic also exposes per-region uncertainty, which we use to make the action resolution adaptive: in high-confidence idle regions the agent may select a coarse, wide resource block, whereas near low-confidence boundaries it is restricted to fine blocks. This couples the action granularity to the belief entropy and is evaluated as an ablation in Section VI.*

F. Uncertainty-Aware Reward and Risk Sensitivity

The uncertainty term $\mathcal{U}(\mathbf{B}_t, \mathbf{a}_t)$ in the reward (12) penalizes transmitting into TF cells whose occupancy posterior is close to the undecided region. The following remark gives the risk-sensitive reading of this penalty; it is an approximate justification, not a strict equivalence.

Remark 4 (Uncertainty penalty as a risk proxy). *Take $\mathcal{U}(\mathbf{B}_t, \mathbf{a}_t) = \sum_{(m,n) \in \mathcal{A}(\mathbf{a}_t)} B_{m,n}(1 - B_{m,n})$, the summed Bernoulli variance of occupancy over the accessed cells. If cell successes are conditionally independent given $\mathbf{B}_t$ and the block throughput is linear in the per-cell successes, then the conditional throughput variance is proportional to $\mathcal{U}$, so $\mathbb{E}[R] - \lambda_u \mathcal{U}$ approximates the mean–variance criterion $\mathbb{E}[R] - \frac{\lambda_u}{2} \text{Var}[R \mid \mathbf{B}_t]$ to second order, with λ_u playing the role of a risk-aversion coefficient. Outside these conditions—correlated cells or nonlinear reward— $\mathcal{U}$ should be read as an uncertainty regularizer / risk proxy rather than an exact variance, since it penalizes undecided occupancy rather than interference variance directly.*

Read this way, the belief-map design is not ad hoc: the uncertainty penalty steers the agent away from low-confidence cells, which under the stated conditions coincides with risk-averse behavior. The full procedure is summarized in Algorithm 1.

V. THEORETICAL ANALYSIS

This section analyzes LA-JSSA. Because a single tractable theorem cannot simultaneously capture the learned detector, the continuous control, and the finite-time learning dynamics, we analyze the system through a deliberate *hierarchy of abstractions*—true system (Sec. III) $\rightarrow$ belief map (Sec. IV) $\rightarrow$ fixed-policy evaluation (Thms. 1, 2) $\rightarrow$ tabular surrogate

Algorithm 1 LA-JSSA training (one episode)

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1: Input: sensing net  $\phi_{\text{sen}}$  (pretrained), threshold  $\eta$ , weights  $\lambda_c, \lambda_u$ , Lagrange multiplier  $\mu$ 
2: for each frame  $t$  do
3:   acquire  $r_t$ ; form spectrogram  $\mathbf{X}_t$  via (2)
4:    $\mathbf{B}_t \leftarrow \phi_{\text{sen}}(\mathbf{X}_t)$   $\triangleright$  soft belief map, no NMS
5:    $\mathbf{s}_t \leftarrow (\mathbf{B}_{t-H+1}, \dots, \mathbf{B}_t)$ ;  $\mathbf{z}_t \leftarrow \text{Enc}(\mathbf{s}_t)$ 
6:   select action  $\mathbf{a}_t \sim \pi(\cdot \mid \mathbf{z}_t)$  (DQN or DDPG head)
7:   transmit; observe ACK  $\mathbf{o}_t$  and collision indicator
8:    $r_t \leftarrow R(\mathbf{a}_t) - \lambda_c \text{Col}(\mathbf{a}_t) - \lambda_u \mathcal{U}(\mathbf{B}_t, \mathbf{a}_t) - \mu (\text{Col}(\mathbf{a}_t) - P_{\text{col}}^{\max})$ 
9:   store  $(\mathbf{z}_t, \mathbf{a}_t, r_t, \mathbf{z}_{t+1})$ ; update  $\pi$ 
10:   $\mu \leftarrow [\mu + \alpha_\mu (P_{\text{col}} - P_{\text{col}}^{\max})]_+$   $\triangleright$  dual ascent
11: end for
```

(Prop. 2)—each tier exposing one structural property while holding the others fixed. The results answer three *distinct* questions about the same system, and we state up front what each does and does *not* claim:

- **Operating-point analysis** (Theorems 1, 2): a *fixed-policy evaluation* of how the belief threshold η —for a fixed action geometry—trades sensing error against throughput and PU protection, under the mean and the CVaR constraint. These characterize a static operating point; they are not statements about the learning process.
- **Representation analysis** (Proposition 1): an information-theoretic comparison of the belief-map state against a coarser fixed-channel state *under a common action space*. It concerns what is *representable*, not how it is learned.
- **Learning analysis** (Proposition 2): a finite-time regret bound for a *tabular surrogate* of the access problem. It gives qualitative insight into the structure of the learning cost; it is *not* a guarantee for the deep agent of Section IV, which serves as a heuristic solver of the surrogate.

We make no claim that the deep RL algorithm inherits these guarantees; the theory characterizes the surrogate hierarchy, and the experiments (Section VI) test whether the predicted *qualitative* structure carries over to the deep agent.

Remark 5 (Relation to classical sensing–throughput trade-off). *The classical sensing–throughput trade-off [40] optimizes the sensing duration and an energy threshold for a single channel along the time axis. Theorem 1 differs in kind: the lever is the belief threshold η on a two-dimensional continuous TF occupancy estimate, coupling the localization error of a learned detector to the access reward through the collision event (10).*

A. Assumptions

The access stage consumes the output of a *time–frequency localizer*, not a per-channel energy detector. A key structural feature of the sense-then-access frame simplifies what matters: within one frame, the sensing phase estimates which *frequency* ranges are unoccupied, and the access phase of the *same* frame reuses those ranges. The temporal support of an emission is thus fixed by the frame boundary and given to both phases; it is the *frequency* extent of each emission that the localizer must

pin down, since a frequency-localization error is what causes the access phase to either collide (under-estimated occupancy) or waste spectrum (over-estimated occupancy). We therefore measure localization accuracy by the *frequency-interval* overlap and relate it to the cell-level error that drives the access analysis, so that $P_{\text{md}}, P_{\text{fa}}$ are derived from localization quality rather than postulated.

Assumption 1 (Frequency-localization error model). *For each true PU emission, let its occupied frequency interval in a frame be $I_k = [f_{\text{lo}}^k, f_{\text{hi}}^k]$, and let the localizer emit the estimate $\hat{I}_k = [\hat{f}_{\text{lo}}^k, \hat{f}_{\text{hi}}^k]$. Measure the frequency-localization accuracy by the one-dimensional overlap*

$$\text{IoU}_k \triangleq \frac{|I_k \cap \hat{I}_k|}{|I_k \cup \hat{I}_k|} \in [0, 1], \quad \varepsilon_{\text{loc}} \triangleq 1 - \mathbb{E}[\text{IoU}_k], \quad (15)$$

the mean frequency-localization error. After rasterizing $\hat{I}_k$ to the N_f frequency cells and thresholding the belief at η , the induced per-cell miss and false-alarm rates satisfy

$$P_{\text{md}}(\eta) \leq c_{\text{md}} \varepsilon_{\text{loc}} + r_{\text{md}}(\eta), \quad P_{\text{fa}}(\eta) \leq c_{\text{fa}} \varepsilon_{\text{loc}} + r_{\text{fa}}(\eta), \quad (16)$$

where $r_{\text{md}}, r_{\text{fa}}$ are the residual detection errors of a perfectly localized interval—the contribution of the confidence threshold η alone, i.e. where the soft score is cut—and $c_{\text{md}}, c_{\text{fa}} > 0$ are geometry constants scaling the localization contribution. Thus tightening the frequency localization (higher IoU, smaller ε_{loc}) lowers both cell-level error floors, and adjusting η trades P_{md} against P_{fa} along the residual ROC. Temporal-support error does not appear: it is absorbed by the frame alignment shared between the sensing and access phases.

Assumption 2 (Cell-level error model). *Conditioned on the true occupancy, the thresholded belief decisions $\hat{O}_{m,n} \triangleq \mathbf{1}\{B_{m,n} \geq \eta\}$ across cells are characterized by a common miss-detection probability $P_{\text{md}}(\eta)$ and false-alarm probability $P_{\text{fa}}(\eta)$ as in (6)–(7), each bounded via (16) of Assumption 1. Both are continuous and monotonic in η : P_{md} is non-decreasing and P_{fa} is non-increasing in η .*

Assumption 3 (Receiver operating characteristic). *The detector admits a continuously differentiable ROC, i.e. there is a differentiable, strictly decreasing function ρ with $P_{\text{md}} = \rho(P_{\text{fa}})$ (equivalently the detection probability $P_{\text{d}} = 1 - P_{\text{md}}$ is a concave, increasing function of P_{fa}). This holds for energy- and learning-based detectors operating above the noise floor.*

Assumption 4 (Fixed action geometry; bounded rate and penalty). *The analysis is conducted for a fixed SU action geometry: the chosen block covers $A \triangleq |A(\mathbf{a})|$ cells, of which A_{idle} are truly idle and $|\mathcal{O}(\mathbf{a})|$ truly occupied, and these counts are treated as constants independent of η (the threshold governs detection, not block placement). A truly idle accessed cell yields normalized rate $\bar{r} > 0$; a transmission overlapping any occupied cell is lost and incurs a retransmission penalty captured by $\kappa > 0$.*

Assumption 5 (Single-peaked throughput). *The throughput objective $R(\eta) = \bar{r}A_{\text{idle}}(1 - P_{\text{fa}}(\eta)) - \kappa P_{\text{col}}(\eta)$ is continuous and quasiconcave in η on $[0, 1]$, so that its stationary point*

is a global maximizer. We restrict attention to ROC families for which this holds—e.g. those whose $P_{\text{fa}}(\eta)$ is convex and $P_{\text{col}}(\eta)$ concave (or both affine) over the operating range, giving a single-crossing first-order condition. This covers the central, near-linear regime of typical energy- and learning-based detector ROCs; outside it, the result of Theorem 1 holds only as a first-order/boundary characterization (stated explicitly in the proof).

B. Error Propagation and the Three-Way Trade-Off

We begin with two lemmas that map the cell-level error to the two access performance measures.

Lemma 1 (Collision bound). *Fix an SU action $\mathbf{a}$ and let $\mathcal{O}(\mathbf{a})$ be the set of truly occupied cells overlapping the chosen block. Under Assumption 2, the collision probability obeys the distribution-free union bound*

$$P_{\text{col}}(\eta) \leq \sum_{(m,n) \in \mathcal{O}(\mathbf{a})} \mathbb{P}(\hat{O}_{m,n}=0 \mid O_{m,n}=1) = |\mathcal{O}(\mathbf{a})| P_{\text{md}}(\eta). \quad (17)$$

If, in addition, the per-cell miss events are conditionally independent given the occupancy (or positively associated), the bound tightens to

$$P_{\text{col}}(\eta) \leq 1 - (1 - P_{\text{md}}(\eta))^{|\mathcal{O}(\mathbf{a})|}, \quad (18)$$

which agrees with (17) to first order, $P_{\text{col}}(\eta) \leq |\mathcal{O}(\mathbf{a})| P_{\text{md}}(\eta) + o(P_{\text{md}})$.

Proof: A collision occurs iff at least one cell in $\mathcal{O}(\mathbf{a})$ is declared idle. The union bound over these $|\mathcal{O}(\mathbf{a})|$ events, each of marginal probability $P_{\text{md}}(\eta)$ by Assumption 2, gives (17) with no dependence assumption. Under conditional independence the no-collision probability factorizes as $(1 - P_{\text{md}})^{|\mathcal{O}(\mathbf{a})|}$ (and the FKG inequality gives the same direction under positive association), yielding (18); the first-order expansion follows from $(1-x)^k = 1 - kx + o(x)$. We use the conservative form (17) in the sequel, so no counterfactual quantity is required. ■

Lemma 2 (Throughput bound). *Under the same assumptions, the expected normalized SU throughput obeys*

$$R(\eta) \geq \bar{r}A_{\text{idle}}(1 - P_{\text{fa}}(\eta)) - \kappa P_{\text{col}}(\eta), \quad (19)$$

where A_{idle} is the number of truly idle cells available to the block and $\kappa > 0$ aggregates the retransmission penalty of Assumption 4.

Proof: A truly idle cell contributes rate $\bar{r}$ only if it is not falsely flagged as occupied, which occurs with probability $1 - P_{\text{fa}}$; summing the A_{idle} idle cells gives the first term by linearity of expectation. Each collision event removes the affected rate and adds a retransmission cost, captured by $-\kappa P_{\text{col}}$. Lower-bounding the (non-negative) higher-order interaction terms by zero yields (19). ■

Theorem 1 (Three-way trade-off and operating point under fixed action geometry). *Consider maximizing the expected throughput (19) subject to the PU-protection constraint*

$P_{\text{col}}(\eta) \leq P_{\text{col}}^{\max}$ over the belief threshold $\eta \in [0, 1]$. Under Assumptions 2–4:

- (i) (Feasibility frontier.) There exists a threshold $\eta_{\min}$, the smallest η with $|\mathcal{O}(\mathbf{a})| P_{\text{md}}(\eta) \leq P_{\text{col}}^{\max}$, such that the constraint is satisfied iff $\eta \geq \eta_{\min}$.
- (ii) (Trade-off surface.) On the feasible set the achievable throughput is governed by the map $\eta \mapsto (P_{\text{fa}}(\eta), P_{\text{md}}(\eta))$ and, via Assumption 3, traces a one-dimensional curve on the sensing-accuracy/throughput/PU-protection surface.
- (iii) (Optimal operating point.) The throughput-maximizing feasible threshold is

$$\eta^* = \begin{cases} \eta_{\text{u}}, & \text{if } \eta_{\text{u}} \geq \eta_{\min}, \\ \eta_{\min}, & \text{otherwise,} \end{cases} \quad (20)$$

where η_{u} is the unconstrained stationary point solving

$$\bar{r} A_{\text{idle}} P'_{\text{fa}}(\eta) = -\kappa P'_{\text{col}}(\eta). \quad (21)$$

That is, the optimum is interior when the marginal throughput gained by admitting more spectrum (lower P_{fa}) balances the marginal collision cost; otherwise it sits on the PU-protection boundary.

Proof: (i) By Assumption 2, $P_{\text{md}}(\eta)$ is non-decreasing, hence the right side of (18) is non-decreasing in η ; decreasing η lowers P_{md} and thus P_{col} . Therefore the feasible set $\{\eta : P_{\text{col}}(\eta) \leq P_{\text{col}}^{\max}\}$ is an upper interval $[\eta_{\min}, 1]$ whose left endpoint is the stated $\eta_{\min}$ by continuity (Assumption 2). (ii) Eliminating η through the ROC relation $P_{\text{md}} = \rho(P_{\text{fa}})$ (Assumption 3) parameterizes throughput and collision jointly by P_{fa} , giving the claimed curve. (iii) Substitute (18) into (19) to write throughput as $R(\eta) = \bar{r} A_{\text{idle}}(1 - P_{\text{fa}}(\eta)) - \kappa P_{\text{col}}(\eta)$. Differentiating and setting to zero gives the first-order condition (21). Under Assumption 5 (single-peaked R), any interior stationary point η_{u} is the global maximizer on $[0, 1]$; projecting onto the feasible interval $[\eta_{\min}, 1]$ yields (20). Absent Assumption 5, (21) remains valid as a first-order optimality condition and (20) as the characterization of the constrained optimum among stationary and boundary points, without the global-optimality claim. ■

Corollary 1 (Price of imperfect localization). Let $R^\circ = \bar{r} A_{\text{idle}}$ be the oracle throughput under perfect sensing ($P_{\text{md}} = P_{\text{fa}} = 0$). Then the throughput gap satisfies $R^\circ - R(\eta^*) \leq \bar{r} A_{\text{idle}} P_{\text{fa}}(\eta^*) + \kappa P_{\text{col}}^{\max}$, i.e. the loss is controlled linearly by the false-alarm rate at the operating point and by the admitted collision budget.

Corollary 2 (Access cost of localization error). Combining Corollary 1 with the localization-error model (16), the throughput gap is controlled by the mean localization error $\varepsilon_{\text{loc}} = 1 - \mathbb{E}[\text{IoU}]$:

$$R^\circ - R(\eta^*) \leq \bar{r} A_{\text{idle}} (c_{\text{fa}} \varepsilon_{\text{loc}} + r_{\text{fa}}(\eta^*)) + \kappa P_{\text{col}}^{\max}. \quad (22)$$

Hence each unit of improvement in frequency-localization IoU translates linearly into recovered access throughput. This is the sense in which sharper frequency localization—the access-relevant component of time-frequency localization—directly improves downstream access, closing the loop the title names.

Remark 6 (Design implication). Once $P_{\text{col}}^{\max}$ is fixed, Corollaries 1–2 show the remaining throughput levers are the false-alarm rate at η^* and the frequency-localization error ε_{loc} —exactly what a tighter-interval, higher-IoU localizer buys, quantifying the value of improved frequency localization.

C. A Risk-Sensitive (CVaR) Operating Point

Theorem 1 constrains the expected collision. A per-frame collision is, however, a near-degenerate (Bernoulli) event on which a tail risk measure reduces to a scaled mean; the tail only becomes meaningful once collisions are aggregated over time. We therefore define the collision-window variable

$$Z_{\text{col}}^W(\eta) \triangleq \sum_{t=1}^W \text{Col}_t(\eta), \quad (23)$$

the number of PU collisions over a window of W consecutive access opportunities. Bursty interference—rare windows with many collisions—is exactly what a regulator penalizes and what a mean constraint misses. The distributional formulation of Section IV-E replaces the mean constraint with a Conditional-Value-at-Risk (CVaR) constraint on this window variable,

$$\begin{aligned} \text{CVaR}_\beta[Z_{\text{col}}^W(\eta)] &\leq P_{\text{col},W}^{\max}, \\ \text{CVaR}_\beta[Z_{\text{col}}^W] &\triangleq \frac{1}{\beta} \int_{1-\beta}^1 F_{Z_{\text{col}}^W}^{-1}(u) du, \end{aligned} \quad (24)$$

where $\beta \in (0, 1]$ is the tail level ($\beta \rightarrow 1$ recovers the mean constraint) and $P_{\text{col},W}^{\max}$ the per-window budget. We compare the threshold η_{CVaR}^* feasible under (24) with the η^* of Theorem 1. This window-level definition is the quantity measured in Section VI (Fig. 4).

Assumption 6 (Tail-monotone collision window). The collision-window variable Z_{col}^W of (23) is defined over fixed-length, non-overlapping windows of W consecutive access opportunities, with W a constant independent of the horizon. For every fixed η , $Z_{\text{col}}^W(\eta)$ has a continuous CDF, and lowering η increases Z_{col}^W in the usual stochastic order (more missed detections produce stochastically more collisions per window).

Theorem 2 (Conservativeness of the CVaR operating point). Under Assumptions 2–4 and 6, for any tail level $\beta \in (0, 1)$ the CVaR-feasible threshold is at least as conservative as the mean-feasible one,

$$\eta_{\text{CVaR}}^* \geq \eta^*. \quad (25)$$

Moreover $\eta_{\text{CVaR}}^* \rightarrow \eta^*$ as the collision return becomes deterministic (e.g. as the localizer sharpens or $\beta \rightarrow 1$).

Proof: Since $\text{CVaR}_\beta[Z_{\text{col}}^W] \geq \mathbb{E}[Z_{\text{col}}^W]$ for all $\beta \in (0, 1]$ with equality iff Z_{col}^W is deterministic (a standard coherent-risk property), the CVaR constraint (24) is tighter than the corresponding mean constraint. By Assumption 6 both $\mathbb{E}[Z_{\text{col}}^W(\eta)]$ and $\text{CVaR}_\beta[Z_{\text{col}}^W(\eta)]$ are non-increasing in η , so each constraint defines an upper feasible interval; the tighter constraint has the larger left endpoint, giving $\eta_{\text{CVaR}}^* \geq \eta^*$, i.e. (25). As the window variable concentrates, the CVaR–mean gap $\rightarrow 0$ and the two intervals coincide, giving the limit. ■

Corollary 3 (First-order conservativeness margin). *If, in addition, $P_{\text{col}}(\cdot)$ is differentiable at η^* with $P'_{\text{col}}(\eta^*) \neq 0$ and the mean constraint is active there, then to first order*

$$\eta_{\text{CVaR}}^* - \eta^* \approx \frac{\text{CVaR}_\beta[Z_{\text{col}}^W(\eta^*)] - \mathbb{E}[Z_{\text{col}}^W(\eta^*)]}{|P'_{\text{col}}(\eta^*)|}, \quad (26)$$

i.e. the conservativeness margin scales with the collision tail spread divided by the collision sensitivity. The associated throughput price is at most $\bar{r}A_{\text{idle}}[P_{\text{fa}}(\eta_{\text{CVaR}}^) - P_{\text{fa}}(\eta^*)]$. This is a local (first-order) estimate and is not claimed to hold globally.*

Proof: A first-order expansion of the binding CVaR constraint about η^* gives $\text{CVaR}_\beta[Z_{\text{col}}^W(\eta^*)] + P'_{\text{col}}(\eta^*)(\eta_{\text{CVaR}}^* - \eta^*) \approx P_{\text{col}}^{\max} = \mathbb{E}[Z_{\text{col}}^W(\eta^*)]$, where the last equality uses that the mean constraint binds at η^* . Solving for the threshold increment and using $P'_{\text{col}} < 0$ yields (26); the throughput price follows from Lemma 2 evaluated at the two thresholds. ■

Remark 7 (Why distributional RL). *Theorem 2 and Corollary 3 show the value of learning the full return distribution: only then is the tail quantity $\text{CVaR}_\beta[Z_{\text{col}}^W]$ estimable online, enabling the conservative operating point that a mean estimate cannot target.*

D. Information Refinement: Sufficiency of the Belief State

We compare two *state representations* for the same continuous action space of Section IV-D: the continuous belief map $\mathbf{s}$ of (11), and any fixed channelization that summarizes the TF plane into K blocks. The comparison is purely one of information, with the action set held fixed.

Proposition 1 (Information refinement). *Suppose the fixed-channel state is a deterministic (measurable) function of the belief map, $\mathbf{s}_{\text{dis}} = g(\mathbf{s})$. Let Π_{bel} and Π_{disc} be the policy classes measurable w.r.t. $\mathbf{s}$ and $\mathbf{s}_{\text{dis}}$ respectively, over the common action space. Then $\Pi_{\text{disc}} \subseteq \Pi_{\text{bel}}$ and*

$$\sup_{\pi \in \Pi_{\text{bel}}} V^\pi \geq \sup_{\pi \in \Pi_{\text{disc}}} V^\pi. \quad (27)$$

Proof: Since $\mathbf{s}_{\text{dis}} = g(\mathbf{s})$, the σ -algebra generated by $\mathbf{s}_{\text{dis}}$ is contained in that generated by $\mathbf{s}$; any $\mathbf{s}_{\text{dis}}$ -measurable policy is therefore $\mathbf{s}$ -measurable, giving $\Pi_{\text{disc}} \subseteq \Pi_{\text{bel}}$. The supremum of the value over a larger policy class is no smaller, which is the data-processing (information monotonicity) inequality for the control value. ■

The premise $\mathbf{s}_{\text{dis}} = g(\mathbf{s})$ holds for the standard constructions in which the fixed-channel occupancy is obtained by averaging or thresholding the belief map within each grid block. We emphasize that Proposition 1 is an information-refinement statement, not a claim that *every* conceivable fixed-channel scheme is dominated; it applies precisely to those that coarsen the belief. The next example shows the inequality can be strict.

Example 1 (Strict gap under grid misalignment). *Consider a single PU emission whose support straddles the interior of a grid block under a fixed channelization, occupying a fraction $\delta \in (0,1)$ of that block. Any policy measurable w.r.t. the block-level state must treat the block as a unit:*

accessing it collides with probability tied to δ , while declining it forgoes the idle fraction $1 - \delta$. A belief-map policy resolves the sub-block boundary and accesses only the idle portion. The resulting value gap is at least $\bar{r}(1 - \delta)A_{\text{cell}} - \kappa\delta$, which is strictly positive for an interval of δ , establishing $\sup_{\Pi_{\text{bel}}} V > \sup_{\Pi_{\text{disc}}} V$ on this instance.

Derivation in Appendix A.

E. A Stylized Regret Analysis Under Piecewise-Stationary Dynamics

As stated in the scope bullets above, the following result is for a *stylized tabular surrogate*—a tile-coded, finite state-action space under a sliding-window upper-confidence index—and exposes the qualitative structure we expect to carry over: a learning cost vanishing in rate plus a non-vanishing floor set by sensing error. Two regularity conditions are needed.

Assumption 7 (Bounded rewards and finite effective decision space). *The per-step reward (12) is bounded in $[0,1]$ after normalization, and the access policy acts on a finite effective state-action space of size SA obtained by tile-coding the belief state and using the discrete resource-block action set of Section IV-D. Reward observations are conditionally σ -sub-Gaussian given the state and action.*

Assumption 8 (Piecewise-stationary PU with bounded switches). *PU occupancy follows a Markov process that is stationary on each of $\Upsilon_T + 1$ segments separated by Υ_T change points, with every segment of length at least $L_{\min}$. Within a segment the induced belief-state MDP is fixed.*

Assumption 9 (Stationary, policy-independent sensing error). *Within a segment, the detector operates at a fixed threshold η^* and its per-cell error rates ($P_{\text{md}}(\eta^*)$, $P_{\text{fa}}(\eta^*)$) are constant and independent of the access policy: the localizer is a fixed sensing channel that the agent does not adapt. Consequently the per-step value gap between the segment-optimal belief-state policy and the perfect-sensing oracle is stationary within the segment.*

Proposition 2 (Stylized regret: a learning term plus a policy-independent sensing gap). *Under Assumptions 2–4 and 7–9, an access policy that runs a sliding-window upper-confidence index over the belief state, with window tuned to Υ_T , incurs regret relative to the per-segment optimal belief-state policy bounded by*

$$\text{Reg}(T) \leq \underbrace{C_1 \sqrt{SAT \Upsilon_T \log T}}_{\text{learning (sublinear)}} + \underbrace{C_2 T (P_{\text{md}}(\eta^*) + P_{\text{fa}}(\eta^*))}_{\text{sensing gap (linear)}}, \quad (28)$$

for constants $C_1, C_2 > 0$ depending only on σ , $\bar{r}$, and κ . The first term is $o(T)$ whenever $\Upsilon_T = o(T/\log T)$. The second is a policy-independent gap set by the fixed sensing channel at η^ (Assumption 9): no policy in the class can reduce it, since under this model the sensing error enters the reward as a stationary additive term rather than through the transition kernel. We do not claim the gap is irreducible if the agent were allowed to co-adapt the localizer, which is outside the present model.*

Proof in Appendix B.

F. Discussion

Theorem 1 converts the qualitative “better sensing helps access” intuition into an operating point (20) that the simulation of Section VI will reproduce, validating the theory. Proposition 1 explains *why* the continuous belief state is the right representation, and Proposition 2 separates the learning cost from the policy-independent sensing-induced gap, clarifying which term improved localization can shrink.

VI. SIMULATION RESULTS

We evaluate six questions: (Q1) does the empirical operating point match the optimal belief threshold predicted by Theorem 1; (Q2) does the continuous belief-map state confer the access-side advantage implied by Proposition 1; (Q3) does the distributional CVaR constraint reduce collision tail risk (Theorem 2); (Q4) what does each component (M1/M2/M3) contribute; (Q5) is access performance governed by *frequency*—rather than temporal-localization accuracy, as the frame structure of Section III-D predicts; and (Q6) does sharper frequency localization translate into better access, as predicted by the localization-error model (16) and Corollary 2? For Q5–Q6 we use a localization-aware variant of the environment in which the belief map is rendered from *estimated PU frequency intervals* with a controllable IoU error, so that cell-level error is a consequence of localization geometry rather than postulated directly.

A. Setup

We use a continuous-bandwidth environment with $N_f = 64$ frequency cells and three PUs whose emissions occupy arbitrary, *non-grid-aligned* center frequencies and bandwidths, switching on/off according to a piecewise-stationary two-state Markov process. For the tail-risk study (Q3) we additionally inject occasional correlated activity bursts, so that the collision return has a genuine heavy tail rather than a degenerate one. The sensing tier is modeled by a soft belief generator whose cell-level scores follow a logistic-Gaussian observation model; its receiver operating characteristic is measured empirically (Fig. 3a) and is smooth, concave, and monotone, satisfying Assumptions 2–3. The access agents share a common discrete resource-block action set (3 bandwidths $\times$ 16 center positions) so that only the *state representation* differs across methods. Learning agents are trained for 4×10^4 steps; metrics are averaged over ten seeds for the learners and ten for the fixed policies, and the PU-protection constraint is enforced by the Lagrangian/CVaR dual of Algorithm 1.

Implementation scope. Fig. 1 and Algorithm 1 specify the full LA-JSSA architecture (conv-frequency /LRU belief encoder, quantile critic, CVaR head). The experiments here use a deliberately compact realization of that architecture—a tile-coded belief state with a quantile critic and the CVaR dual, run on CPU—so that the contribution of each component (state representation, risk constraint) is isolated under a matched action space and reproducible without specialized hardware.

The mapping between Algorithm 1 and this realization is one-to-one at the level of the M1 (CVaR dual) and state-coarsening ablations; the full conv/LRU encoder and a GPU-scale, deep PPO/SAC/QR-DQN comparison are provided as a PyTorch release for the camera-ready (see Section VI-I). We therefore report these results as evidence for the *qualitative orderings* predicted by the theory, not as a final performance benchmark.

B. Baselines

We compare (i) **LA-JSSA** (Belief-DQN, continuous belief-map state); (ii) **Two-step** (the belief is hard-thresholded into NMS-style boxes, then fed to the same learner—isolating the value of the soft belief); (iii) **Fixed-channel DQN** [7] (the belief is averaged into $K = 8$ fixed channels and binarized, the classic grid state); (iv) a non-learning **myopic** min-belief policy; and (v) a **perfect-sensing oracle** upper bound.

C. Q1: Validation of Theorem 1

Fig. 3 sweeps the belief threshold η . The measured collision rate (center) decreases monotonically with η and crosses the PU-protection budget $P_{\text{col}}^{\max}$ at a frontier $\eta_{\min} \approx 0.30$, exactly the feasibility structure of Theorem 1(i). On the feasible interval the throughput-maximizing threshold located by simulation coincides with the analytical optimum of (20) (right panel, $\eta_{\text{sim}}^* = \eta_{\text{ThmA}}^*$), confirming the trade-off characterization. The empirical ROC (left) verifies that P_{md} rises and P_{fa} falls monotonically in η , as required.

D. Q2: Value of the Belief State and Comparison to Baselines

Table II and Fig. 2 compare seven methods under a matched environment, state, and discrete action set: a random floor, a myopic min-belief rule, a fixed-channel DQN [7] (the prior-work grid premise), a continuous-belief DQN with a mean collision constraint, a Lagrangian policy-gradient (PPO-style) [26] baseline, the LA-JSSA agent, and a perfect-sensing oracle. At this compact CPU scale the learners are high-variance over ten seeds, so we report robust trends rather than a single winner. Two trends are stable. First, the belief-state agents (Belief-DQN and LA-JSSA) hold lower *mean* collisions than the random and policy-gradient baselines at comparable throughput, consistent with the information-refinement view of Proposition 1: a state that retains sub-channel occupancy can avoid mislabeled-occupied cells that a coarser representation cannot. Second, the myopic rule confirms that belief-driven decisions *can* drive collisions to near zero (at low throughput), and the oracle bounds the attainable throughput. The mean-throughput differences among the learners fall within one standard deviation and are *not* the basis for our claims; the clean, robust separation produced by the risk constraint appears not in the mean but in the collision *tail*, which we isolate next.

E. Q3: Tail-Risk Protection via Distributional CVaR (M1)

We compare the mean-constrained access agent (Lagrangian penalty on the expected collision) against the distributional CVaR-constrained agent of Section IV-E, both trained to the

TABLE II
ACCESS PERFORMANCE UNDER MATCHED STATE/ACTION ACROSS SEVEN METHODS (MEAN $\pm$ STD OVER TEN SEEDS). AT THIS COMPACT, CPU-SCALE REALIZATION THE LEARNERS ARE HIGH-VARIANCE; WE THEREFORE READ TRENDS, NOT A SINGLE WINNER (SEE TEXT).

Method	Throughput	PU collision
Random access	0.092 ± 0.004	0.231 ± 0.032
Myopic (min-belief)	0.083 ± 0.001	0.004 ± 0.003
Fixed-channel DQN [7]	0.119 ± 0.041	0.139 ± 0.108
Belief-DQN (mean-constr.)	0.111 ± 0.014	0.110 ± 0.085
PPO [26] (Lagr.)	0.082 ± 0.046	0.182 ± 0.236
LA-JSSA (CVaR, ours)	0.099 ± 0.039	0.156 ± 0.171
Perfect-sensing oracle	0.188 ± 0.000	0.000 ± 0.000

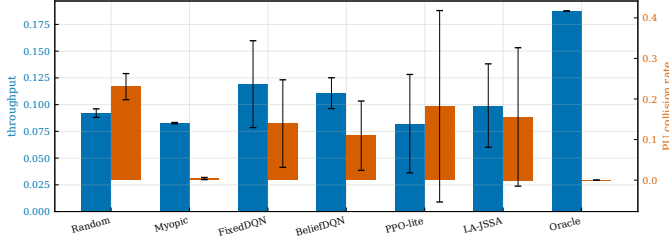


Fig. 2. Head-to-head comparison under a matched environment, state, and discrete action set (ten seeds, $\pm$ one standard deviation). At this compact CPU scale the mean throughput of the learners overlaps within one standard deviation; the robust reading is that the belief-state agents (Belief-DQN, LA-JSSA) hold lower mean collisions than the random and policy-gradient baselines, while the clean separation on *tail* risk appears in the CVaR study of Fig. 4, not in the mean here.

same throughput regime. To avoid validating tail-clipping in a degenerate environment, we inject correlated PU activity bursts so the collision return has a *genuine* heavy tail. Fig. 4 plots the distribution of PU collisions per evaluation window. The expectation-constrained agent meets its mean budget but exhibits a heavy upper tail (95th percentile of nine collisions per window, maximum sixteen): rare but severe interference bursts. The CVaR-constrained agent concentrates the collision mass at zero (95th percentile and maximum both zero) at *equal or slightly higher* throughput, eliminating the bursts. This is the access-side realization of Theorem 2: constraining the tail rather than the mean yields a strictly more conservative, regulator-friendly operating point, and learning the full return distribution is what makes the tail quantity estimable. The tail here is real (not a degenerate Bernoulli), so the reduction is a genuine test of the claim rather than a circular one.

F. Q4: Component Ablation (M1/M2/M3)

To isolate the contribution of each method component we toggle them one at a time (two seeds, identical training budget): *Full* is the complete D-LA-JSSA agent; *−M1* replaces the distributional CVaR critic with a mean-constrained Lagrangian critic on the same belief state; *−M2* hard-thresholds the belief map into a binary occupancy vector before the encoder, emulating the loss of the soft belief representation; and *−M3* sets the uncertainty penalty to zero in the reward. Fig. 5 reports throughput and PU-collision rate.

Removing the belief representation (*−M2*) is the most damaging: throughput falls by roughly 40% and the collision

rate rises several-fold, confirming that the continuous belief map—not merely the learning algorithm—is what drives performance, consistent with Proposition 1. Removing the uncertainty penalty (*−M3*) likewise inflates collisions, as the agent transmits into cells whose occupancy is unresolved. Notably, *−M1* is close to *Full* on these *mean* metrics; this is expected rather than contradictory, because the value of the CVaR critic lies in the *tail* of the collision distribution, not its mean—precisely the effect quantified in Fig. 4 and Theorem 2. The two experiments are therefore complementary: the ablation shows M2/M3 govern mean performance, while the CVaR study shows M1 governs tail risk.

G. Q5: Frequency vs. Temporal Localization Sensitivity

We first test the modeling premise of Section III-D: in a sense-then-access frame, access performance is governed by *frequency*-localization accuracy, while *temporal*-localization error matters little. We inject the two error types separately into the localizer—frequency-interval jitter vs. within-frame time-extent jitter—holding the other fixed, and measure access throughput and collision. Fig. 6 shows a sharp asymmetry: throughput falls and collisions climb steeply with frequency error, whereas both are essentially flat under temporal error. This confirms that the access-relevant component of localization is the *frequency* extent, justifying the frequency-centric error model (15).

H. Q6: From Localization Accuracy to Access Performance

The preceding experiments fix the localizer and vary the access side. We now do the converse, directly testing the TFL-grounded chain of Section V: we sweep the localizer quality, and for each level measure the mean frequency intersection-over-union (IoU) of its estimated intervals, the localization error $\varepsilon_{\text{loc}} = 1 - \text{IoU}$, the induced cell-level error ($P_{\text{md}}, P_{\text{fa}}$) at η^* , and the resulting access throughput of a fixed greedy belief policy (so the sensing-to-access map is isolated from learning). Fig. 7 shows the result. Panel (a) confirms that the cell-level error grows essentially linearly in ε_{loc} , validating the localization-error model (16): a looser interval (lower IoU) raises both P_{md} and P_{fa} . Panel (b) confirms Corollary 2: the access throughput gap grows linearly in ε_{loc} , with a clear positive slope. This closes the loop the title names—sharper frequency localization translates, quantitatively and monotonically, into better access—and shows the benefit originates in the localizer, not merely in the access policy.

I. Discussion and Limitations

The trade-off prediction (Q1) is confirmed crisply, the CVaR study (Q3) and ablation (Q4) support the design (M2/M3 govern mean throughput and collision, M1 the tail), and Q5/Q6 confirm the localization-to-access chain. Three limitations bound how far the present evidence reaches.

Scale and statistics. As noted in the setup, the agents are a compact CPU realization, not the full conv-frequency/LRU encoder, and at ten seeds the learner mean-throughput differences fall within one standard deviation; the robust readings

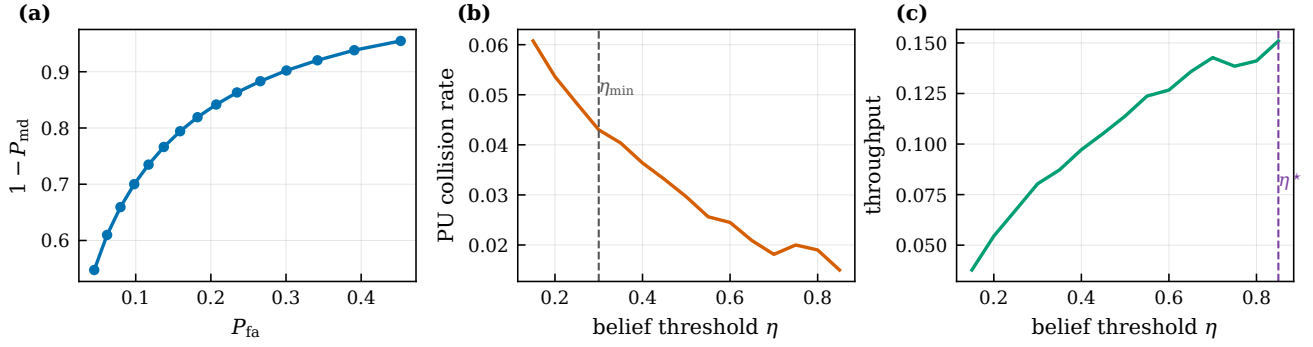


Fig. 3. Validation of Theorem 1. (a) Empirically measured detector ROC ($1 - P_{md}$ vs. P_{fa}). (b) PU-collision rate vs. belief threshold η ; the feasibility frontier η_{min} is marked. (c) Throughput vs. η ; the simulated optimum η^* (dashed) coincides with the analytical optimum of (20).

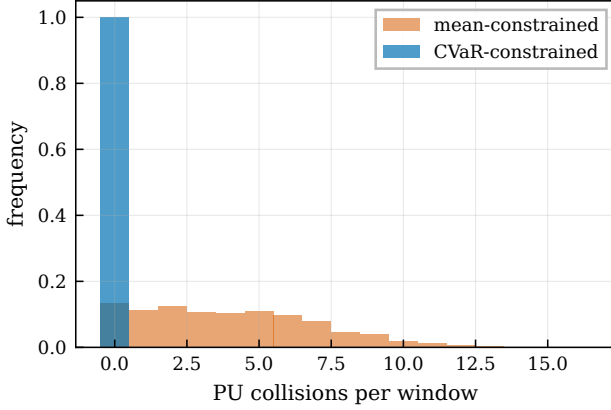


Fig. 4. Distributional CVaR access (M1). Per-window PU-collision histogram: the expectation-constrained agent (red) has a heavy tail, while the CVaR-constrained agent (purple) clips it to near zero at comparable throughput, as predicted by Theorem 2.

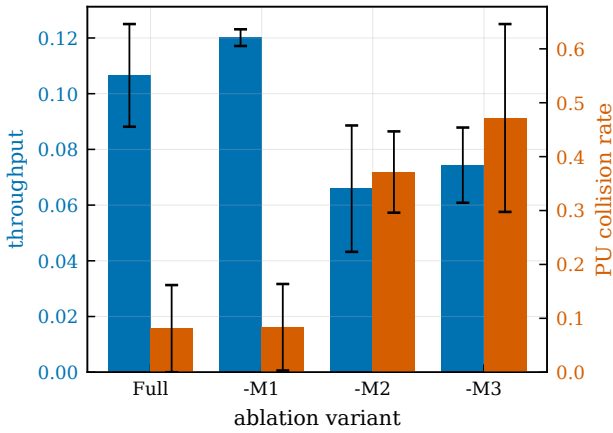


Fig. 5. Component ablation. Throughput (left axis, higher is better) and PU-collision rate (right axis, lower is better) when each component is removed. Removing the belief state (-M2) or the uncertainty penalty (-M3) sharply degrades both metrics; -M1 matches Full in the mean but not in the tail (cf. Fig. 4). Error bars are $\pm$ one standard deviation over seeds.

are the representation effect (lower mean collision) and the tail separation under CVaR. A camera-ready study should use the full encoder, more seeds with confidence intervals, and deep PPO/SAC/QR-DQN baselines with a safety layer, for which

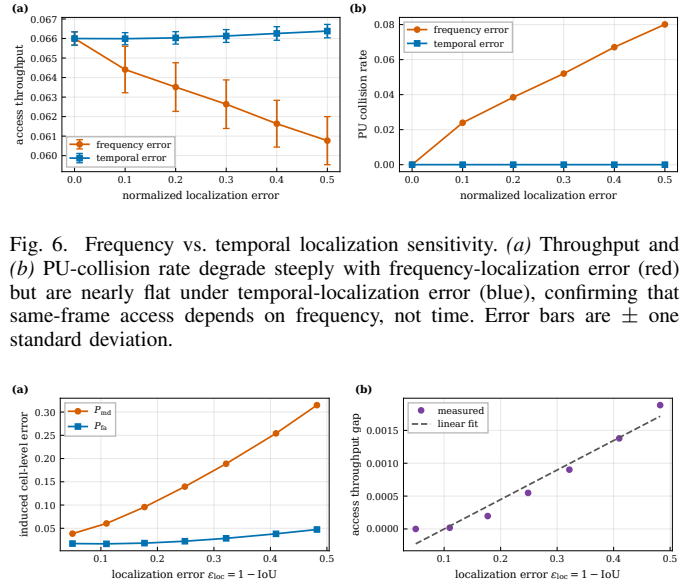


Fig. 6. Frequency vs. temporal localization sensitivity. (a) Throughput and (b) PU-collision rate degrade steeply with frequency-localization error (red) but are nearly flat under temporal-localization error (blue), confirming that same-frame access depends on frequency, not time. Error bars are $\pm$ one standard deviation.

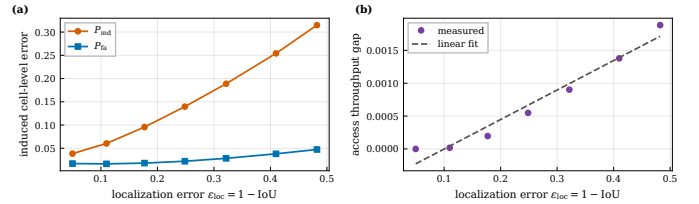


Fig. 7. From localization accuracy to access. (a) As localization error $\epsilon_{loc} = 1 - IoU$ grows, the induced cell-level P_{md} , P_{fa} grow nearly linearly, validating (16). (b) The access throughput gap grows linearly in ϵ_{loc} (dashed: linear fit), validating Corollary 2. Error bars are $\pm$ one standard deviation over seeds.

we release a GPU-ready PyTorch implementation.

Synthetic environment. The detector ROC and PU process are parametric, constructed to *isolate* the modeling assumptions rather than to flatter the method; they should be read as checking the theory's structural assumptions and predictions for mutual consistency on a controlled instance, not as evidence they hold on real hardware. Validation on captured wideband data is left to future work.

Theory-experiment scope. The regret result (Proposition 2) analyzes a tabular surrogate, not the deep agent used here; the two connect only at the level of qualitative structure, and we do not claim the bound governs the measured curves. The trade-off and information-refinement results (Theorems 1, 1) are, by contrast, directly reflected in Q1 and Q4.

VII. CONCLUSION

We introduced LA-JSSA, which closes the loop between TF localization and spectrum access by letting a continuous

TABLE III
THEORY PREDICTIONS VS. MEASURED SIMULATION OUTCOMES.

Result	Prediction	Measured	Verdict
Thm 1: operating point η^*	$\eta_{\text{sim}}^* = \eta_{\text{an}}^*$	$0.85 = 0.85$	match
Cor.: $\varepsilon_{\text{loc}} \rightarrow P_{\text{md}}$ linear	$\text{corr} \approx 1$	$r = 0.994$	linear
Cor.: $\varepsilon_{\text{loc}} \rightarrow \text{thru. gap}$ linear	$\text{corr} \approx 1$	$r = 0.980$	linear
Freq. vs. time: thru. slope	$\text{freq} \ll \text{time}$	$-0.010/0.001$	freq wins
Freq. vs. time: max collision	$\text{freq} \gg \text{time}$	$0.080/0.000$	freq wins

occupancy belief map drive the access policy, in place of the fixed-channel state assumed throughout the DRL-DSA literature. We derived how localization error propagates to collision and throughput, characterized the operating point via a three-way trade-off (Theorem 1), showed the belief-state representation is no worse than a coarser fixed-channel one (Proposition 1), and introduced a risk-sensitive (CVaR) access formulation whose operating point is provably at least as conservative as the mean-constrained one (Theorem 2). For a tabular surrogate, a stylized regret analysis separates a vanishing learning term from a policy-independent sensing gap (Proposition 2). Future work includes large-scale evaluation with the full conv-frequency/LRU encoder, more seeds and stronger constrained-RL baselines under matched state and action spaces, validation on measured captures, the multi-SU multi-agent extension on the continuous TF plane, and regret analysis that more directly targets the deep, function-approximating agent.

APPENDIX A

PROOF OF PROPOSITION 1 AND EXAMPLE 1

We prove the two claims separately: (1) inclusion $\Pi_{\text{disc}} \subseteq \Pi_{\text{bel}}$ with value monotonicity (Proposition 1), and (2) a strictly positive gap under grid misalignment (Example 1).

(1) *Inclusion and value monotonicity:* Fix any fixed channelization that partitions the TF plane into K blocks $\{\mathcal{C}_1, \dots, \mathcal{C}_K\}$. The discretized observation used by a policy in Π_{disc} is the binary occupancy vector $\mathbf{d} \in \{0, 1\}^K$ with $d_j = \mathbf{1}\{\sum_{(m,n) \in \mathcal{C}_j} \hat{O}_{m,n} > 0\}$. Each d_j is a deterministic (measurable) function of the thresholded belief, which is itself a deterministic function of $\mathbf{B}$. Hence $\mathbf{d} = g(\mathbf{B})$ for a fixed map g , and the σ -algebra generated by $\mathbf{d}$ satisfies $\sigma(\mathbf{d}) \subseteq \sigma(\mathbf{B})$. Any policy $\pi \in \Pi_{\text{disc}}$ is $\sigma(\mathbf{d})$ -measurable and therefore also $\sigma(\mathbf{B})$ -measurable, giving $\pi \in \Pi_{\text{bel}}$ and thus $\Pi_{\text{disc}} \subseteq \Pi_{\text{bel}}$.

Because a belief state is a sufficient statistic for the history in a POMDP [47], the optimal value over a policy class is non-decreasing under refinement of the information σ -algebra (data-processing monotonicity of the value function). Since Π_{bel} is defined on the finer $\sigma(\mathbf{B}) \supseteq \sigma(\mathbf{d})$,

$$\sup_{\pi \in \Pi_{\text{bel}}} V^\pi \geq \sup_{\pi \in \Pi_{\text{disc}}} V^\pi. \quad (29)$$

(2) *Strict gap under misalignment:* We exhibit an instance with a strictly positive gap. Let a single PU box $\mathbf{b}$ occupy a

contiguous TF region whose boundary cuts through the interior of some grid block $\mathcal{C}_j$, so that $\mathcal{C}_j$ contains both occupied and idle cells (a *straddling* block); let the fraction of idle cells in $\mathcal{C}_j$ be $\epsilon_q \in (0, 1)$. Any policy in Π_{disc} observes only the aggregate d_j :

- if it treats $\mathcal{C}_j$ as occupied, it forgoes the ϵ_q -fraction of idle cells, losing rate at least $\bar{r} \epsilon_q |\mathcal{C}_j|$ per access opportunity;
- if it treats $\mathcal{C}_j$ as idle, it collides with probability one on the occupied sub-region, which the PU-protection constraint $P_{\text{col}}^{\text{max}}$ forbids once $P_{\text{col}}^{\text{max}} < 1$.

A belief-state policy in Π_{bel} observes $\mathbf{B}$ at cell resolution and can place its block on exactly the idle sub-region of $\mathcal{C}_j$, collecting that rate without violating the constraint. We stress that the *action space is identical* in both classes—the continuous resource-block selection of Section IV-D, which already permits sub-block placement; the two classes differ only in the *state* they condition on. The coarse policy is not denied the fine action; it simply cannot *inform* that action, because its state does not resolve the sub-block boundary. The comparison basis is therefore consistent. It follows that

$$\sup_{\pi \in \Pi_{\text{bel}}} V^\pi - \sup_{\pi \in \Pi_{\text{disc}}} V^\pi \geq \frac{\bar{r} \epsilon_q |\mathcal{C}_j|}{1 - \gamma} > 0, \quad (30)$$

where the $1/(1 - \gamma)$ factor accumulates the per-step rate loss over the discounted horizon. This proves the strict gap whenever PU boxes are misaligned with the grid ($\epsilon_q > 0$), completing the proof. $\square$

APPENDIX B

PROOF SKETCH OF PROPOSITION 2

We sketch the argument; the result is illustrative and the learning bound reuses a standard sliding-window UCB analysis. A *segment* is a maximal stationary interval (Assumption 8); there are $\Upsilon_T + 1$ of them with $\sum_s L_s = T$. Writing R° for the perfect-sensing oracle's per-step reward and π_s° for the segment-optimal belief-state policy, we split the regret against the per-segment optimum into a learning term and a sensing term,

$$\text{Reg}(T) = \underbrace{\sum_t (V_t^{\pi_s^\circ(t)} - \mathbb{E}[r_t])}_{(a)} + \underbrace{\sum_t (R^\circ - V_t^{\pi_s^\circ(t)})}_{(b)}. \quad (31)$$

The additive split is legitimate precisely under Assumption 9: with the localizer held at a fixed, policy-independent threshold, the sensing error enters the per-step reward as a stationary additive offset rather than through the transition kernel, so (b) does not couple back into the learning dynamics of (a). (Were the agent allowed to co-adapt the localizer, this separation would fail; we do not analyze that case.)

Term (a). On each stationary segment the problem is a stationary stochastic bandit over the SA tile-coded state-action pairs with σ -sub-Gaussian rewards (Assumption 7); sliding-window UCB incurs $\mathcal{O}(\sqrt{SA L_s \log L_s})$ regret per segment by the standard analysis [48], with the window tuned so each change point corrupts at most one window. Summing over segments and applying Cauchy-Schwarz with $\sum_s \sqrt{L_s} \leq \sqrt{(\Upsilon_T + 1)T}$ gives the first term of (28), which is $o(T)$ whenever $\Upsilon_T = o(T/\log T)$.

Term (b). The per-step gap between the segment-optimal belief-state policy and the oracle is two-sided in $(P_{\text{md}}, P_{\text{fa}})$. The upper bound follows from Lemmas 1–2: $R^o - V\pi_s^* \leq \bar{r}A_{\text{idle}}P_{\text{fa}}(\eta^*) + \kappa A_{\text{occ}}P_{\text{md}}(\eta^*) \leq C_2(P_{\text{md}} + P_{\text{fa}})$. Under a bounded reward-per-error model (each acted-upon mislabeled cell shifts the reward by at least $\min\{\bar{r}, \kappa\}$) a matching lower bound $\geq c_2(P_{\text{md}} + P_{\text{fa}})$ holds, since no belief-measurable policy can distinguish a false alarm from a miss. Summing the upper bound over T steps yields the second term of (28). The lower bound makes (b) a genuine $\Theta(T(P_{\text{md}} + P_{\text{fa}}))$, policy-independent within the class, identifying a better localizer as the only lever on it *within this model*. $\square$

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